

Shift-Share Construction and Empirical Specification

We construct a municipality-year shift-share measure of exposure to destination-specific foreign exchange movements using two ingredients: baseline migrant allocation across destinations and annual destination-level exchange-rate shocks. The construction follows the logic of khanna2025abundance, who combine pre-shock origin-destination exposure weights with destination-level exchange-rate shocks to obtain an origin-level treatment. Our setup differs in one important respect: because we do not observe migrant income by destination, we use baseline migrant *counts* rather than baseline migrant *income* to form the exposure weights.

Let m index municipalities, d index destination countries, and t index years. Let N_{md0} denote the number of baseline migrants from municipality m to destination d in the 2001 Census, $N_{m0} = \sum_d N_{md0}$ denote the total number of baseline migrants from municipality m , and Pop_{m0} denote the baseline municipality population. We define the baseline destination share of migrants as

$$s_{md0} = \frac{N_{md0}}{N_{m0}},$$

and baseline migration intensity as

$$MI_{m0} = \frac{N_{m0}}{Pop_{m0}}.$$

Next, let Z_{dt} denote the destination-year foreign exchange shifter. In the data, this is constructed from the destination-currency-per-Nepali-rupee cross-rate and is measured either as an index relative to 2001 or as cumulative growth since 2001. Using these destination-year shocks, we first define a municipality-level weighted foreign exchange shock,

$$FXshock_{mt} = \sum_d s_{md0} Z_{dt}.$$

This term captures the *composition* of migrant destinations: municipalities whose migrants were more concentrated in destinations experiencing larger exchange-rate movements receive larger values of $FXshock_{mt}$. By construction, the destination shares sum to one within municipality, so this object is a weighted average of destination-level shocks rather than a full exposure measure. This mirrors the weighted-average shock term in khanna2025abundance, which captures how exposed an origin is to destination shocks through the distribution of migrants across destinations.

To obtain the full municipality-year exposure measure, we scale this composition-based shock by baseline migration intensity. Substituting the definitions above yields the equivalent and more transparent expression:

$$SSIV_{mt} = \frac{\sum_d N_{md0} Z_{dt}}{Pop_{m0}}.$$

This is our main shift-share regressor. Intuitively, $SSIV_{mt}$ is larger when a municipality sent more migrants abroad at baseline, and when those migrants were disproportionately concentrated in destinations experiencing larger exchange-rate changes. It therefore combines both the *scale* of migration exposure and the *composition* of migrant destinations. In this sense, it is the municipality-level analog of the incomplete-shares shift-share measure in khanna2025abundance: the weighted-average shock $FXshock_{mt}$ alone does not capture differences in overall exposure, while multiplying by baseline migration intensity converts it into a per-resident measure of municipality-level foreign exchange exposure.

We estimate the following reduced-form specification:

$$y_{mt} = \alpha_m + \gamma_t + \beta SSIV_{mt} + \sum_{\tau \neq t_0} \lambda_\tau (MI_{m0} \times \mathbf{1}\{t = \tau\}) + X'_{m0} \Theta_t + \varepsilon_{mt}, \quad (1)$$

where y_{mt} is the outcome of interest in municipality m and year t , α_m are municipality fixed effects, and γ_t are year fixed effects. Municipality fixed effects absorb all time-invariant differences across municipalities, including persistent differences in migration history, remoteness, or baseline development. Year fixed effects absorb aggregate shocks common to all municipalities in a given year, including national macroeconomic conditions and policy changes.

Our coefficient of interest is β , which measures the reduced-form effect of municipality exposure to destination-specific foreign exchange shocks. Because the exposure measure is constructed using baseline migrant counts, β should be interpreted as the effect of a higher foreign exchange shock faced by baseline migrant networks per resident, rather than the effect of a shock to baseline migrant income per resident as in khanna2025abundance.

The interaction terms between baseline migration intensity and year indicators,

$$\sum_{\tau \neq t_0} \lambda_\tau (MI_{m0} \times \mathbf{1}\{t = \tau\}),$$

play an important role. Since MI_{m0} is fixed in the baseline year, it does not vary over time within a municipality. However, municipalities with higher baseline migration intensity may systematically follow different time paths even in the absence of destination-specific exchange-rate shocks. For example, high-migration municipalities may experience different trends in remittances, labor supply, consumption, or demographic outcomes for reasons unrelated to the particular destinations to which their migrants were exposed. Interacting MI_{m0} with year fixed effects allows these high-migration municipalities to evolve differently over time in a fully flexible way. As a result, identification comes from differences in exposure to destination-specific foreign exchange variation, rather than from the fact that some municipalities simply had more migrants at baseline. This follows the incomplete-shares logic emphasized by khanna2025abundance, where the scale component of exposure must be allowed to have its own time-varying effects.

Finally, $X'_{m0} \Theta_t$ denotes baseline municipality characteristics interacted with year fixed effects. These controls flexibly account for the possibility that mu-

nicipalities with different initial conditions—such as population size, literacy, urbanization, or baseline economic structure—experience different outcome trajectories over time. Including these interactions ensures that the coefficient on $SSIV_{mt}$ is not driven by differential trends correlated with baseline municipality characteristics. Taken together, the specification isolates variation coming from the interaction of predetermined migrant allocation across destinations and destination-specific foreign exchange shocks over time.